The University of Nottingham Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL THREE MODULE, SPRING SEMESTER 2022-2023

Computer Vision

Time allowed: TWO HOURS

Candidates may complete the front cover of this exam and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer ALL THREE Questions in Section A. Answer TWO out of THREE Questions in Section B.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

ADDITIONAL MATERIAL: Nil.

SECTION A

Answer ALL THREE Questions in This Section.

Question 1. The following questions are pertaining to Image Features.

[**overall 13 marks**]

a. Determine the Local Binary Pattern (LBP) for the central pixel in the image patch in Figure 1. Explain in general how LBP features are further organised to produce image features that can be used for downstream tasks, e.g., image classification.

90	95	91
85	92	95
90	100	98

Figure 1

[5 Marks]

b. In Figure 2, an original seed image and its blurred versions are shown from left to right in increasing blurriness by Gaussian filters. The coloured circles in three images are the SIFT feature points detected for each of the seed images.

i. Explain why there are many SIFT points around the edge of the seed body in the original image on the left.

[3 Marks]

ii. Describe the trend of the SIFT points in images from left to right, and explain why it is the case.

[3 Marks]

iii. If the original image is added with Gaussian noise, will it affect the detection of SIFT points? Explain.

[2 Marks]

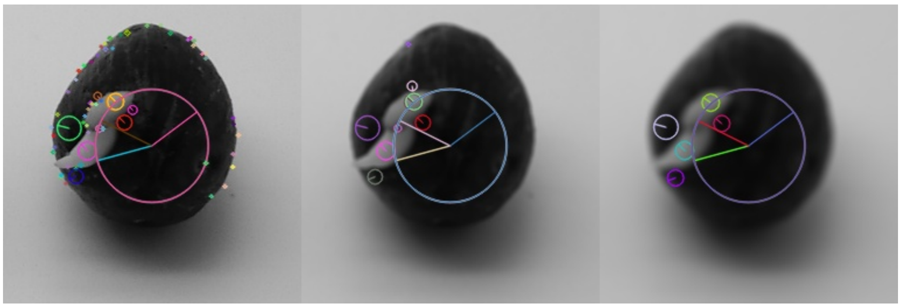


Figure 2

Question 2. The following questions are pertaining to Camera Geometry and Epipolar Geometry.

[**overall 13 marks**]

a. A pinhole camera has a focal length of 35mm, 72dpi (i.e., 28.35 pixels per cm) for both horizontal and vertical resolution, and 4896 pixels horizontally and 3264 pixels vertically. Assuming there is no distortion in the projection, calculate the intrinsic parameter matrix. Show your work.

[7 Marks]

b. Figure 3 shows the epipolar geometry that governs the relations between a scene point P and its projection p in the left image (with camera centre O_1) and p' in the right image (with camera centre O_2). Algebraically, it is characterised by a fundamental matrix **F** which satisfies the equation

$$p^T \mathbf{F} p' = 0 \qquad (1)$$

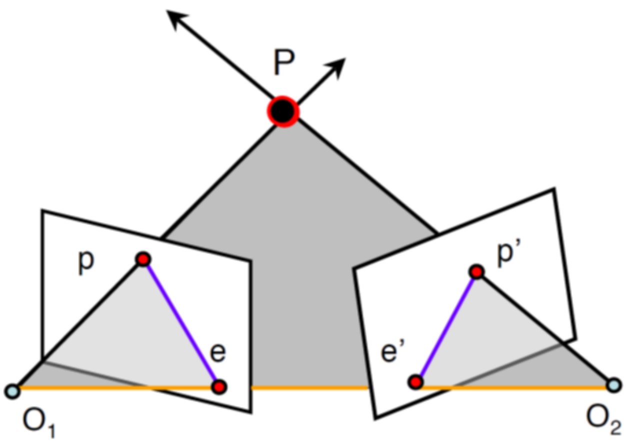


Figure 3

(a) Specify the components that form the fundamental matrix in equation 1. You do not need to specify the operations among the components for the calculation of the fundamental matrix.

[2 Marks]

(b) Explain the concept of epipolar geometry and how it leads to equation 1. You do not need to derive any formula or equations.

[4 Marks]

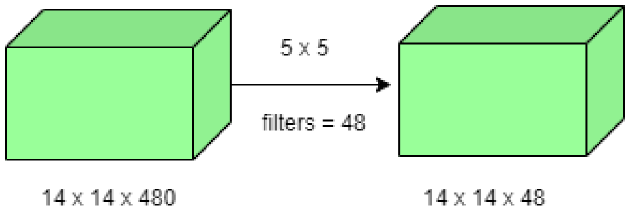
Question 3. The following questions are pertaining to Convolutional Neural Networks.

[**overall 14 marks**]

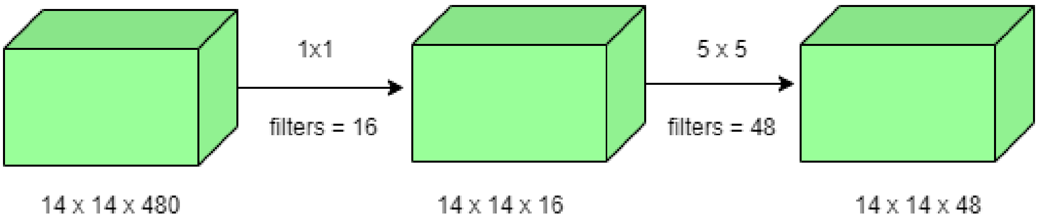
a. Given a 64-channel input feature map of size 128×128, what is the size of the output feature map if you convolve it with a 5×5 filter, without zero padding, and with stride 2, to obtain a 128-channel output feature map? Show your work.

[2 Marks]

b. In the architecture of GoogLeNet, 1×1 convolution is used frequently to improve the efficiency of the network. How does it improve the efficiency of the network? Does it do so by reducing the number of learnable parameters? Explain using the example showing in Figure 4.



(a) Without 1x1 convolution



(b) With 1x1 convolution

Figure 4

[5 Marks]

c. Given an input RGB image of size 256×256, design a simple Convolutional Neural Network (CNN) that consists of three trainable layers to output a feature map of size 64×64 with 256 channels. You may only choose from operations of 3×3 convolution, 1×1 convolution, and/or 2×2 max pooling. Explain your design.

[7 Marks]

SECTION B

Answer TWO Out of THREE Questions in This Section.

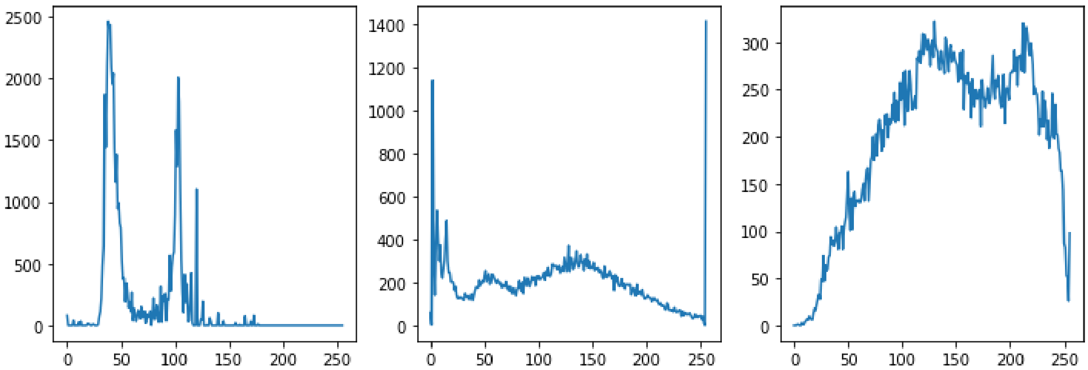
Question 4: The following questions are pertaining to Image Segmentation and Optical Flow.

[**overall 30 marks**]

a. A natural scene image is shown in Figure 5. The Hue, Saturation, and Value (HSV) channels of the image are shown in Figure 6a from left to right, and the histograms of the three channels are shown in Figure 6b, respectively. You are required to use Gaussian Mixture Model (GMM) to segment the image into three regions, i.e., water fall, hill, and sky, based on the HSV representation of the image. Describe and discuss your design, making use of the histograms.

Figure 5

(a) HSV



(b) Histograms of HSV channels

Figure 6

[16 Marks]

b. Two 5×5 image patches at time t and $t+1$ are given in Figure 7(a) and (b), respectively. Calculate the image gradients along x, y, and the time t directions for a 3×3 window centered at the central pixel of the image patch in (a). Assuming the origin of image coordinate system is at the upper-left corner of the image, use the formula $I_x = I(x+1, y) - I(x, y)$ and $I_y = I(x, y+1) - I(x, y)$ to calculate the image gradients along x, y axis, respectively. Based on Lucas-kanade method, explain how these image gradients may be combined to estimate the optical flow vector of the pixel at the centre of the image patch in (a). There is no need to calculate the optical flow vector.

215	214	214	214	214
215	214	214	214	214
215	215	215	215	214
215	215	215	215	213
215	214	214	214	213

(a)

201	201	200	201	212
201	201	196	196	207
201	200	195	195	200
201	198	192	193	193
201	196	196	189	191

(b)

Figure 7

[14 Marks]

Question 5: The following questions are pertaining to Stereo Vision

[**overall 30 marks**]

a. A pair of rectified stereo images are shown in Figure 8. The Block-Matching algorithm is used to find the dense correspondence of pixels between the left and right images. Discuss at least three (3) problems of block-matching with respect to the stereo images.

Figure 8

[6 Marks]

b. A pair of rectified stereo images are shown in Figure 9. The pixel at the centre of the black circle in the left image corresponds to the pixel at the centre of the black circle in the right image. The pixel coordinates (x, y) are $(109, 22)$ for the left and $(58, 22)$ for the right, respectively. Similarly, the pixel at the centre of the white circle in the left image corresponds to the pixel at the centre of the white circle in the right image. The pixel coordinates are $(117, 45)$ for the left and $(68, 45)$ for the right, respectively.

Explain the concept of 3D cost volume and describe how you will build the 3D cost volume for the pair of images. Use the matching pixels at the centre of the black circles and matching pixels at the centre of the white circles in your demonstration.

(a) Left image (b) Right image

Figure 9

[10 Marks]

c. Figure 10 shows a typical architecture of deep learning based stereo vision model which mimics the traditional stereo vision pipeline. Briefly explain the functionality of each block (A)dense/sparse feature extraction, (B) feature matching, (C) cost volume regularisation, (D) obtain raw disparity map, and (E) post processing and refinement, respectively.

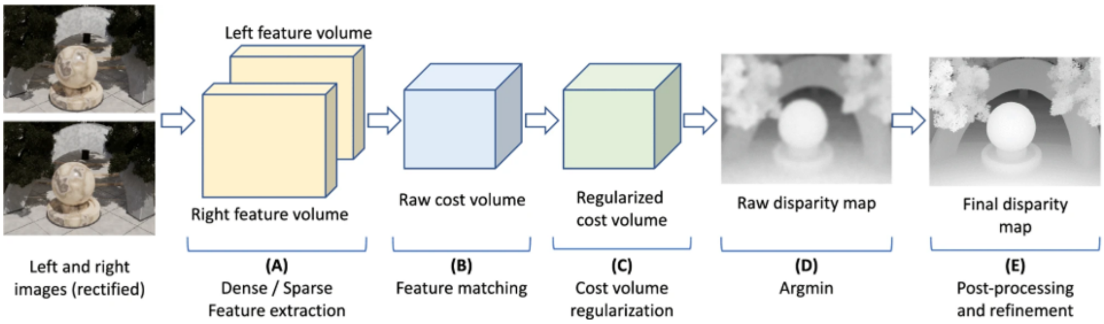


Figure 10

[14 Marks]

Question 6: The following questions are pertaining to Object Recognition and Convolutional Neural Networks.

[**overall 30 marks**]

a. The Viola-Jones method relies on efficient integral images to generate a large amount of simple image features. Given the 3×3 image patch shown in Figure 11, calculate its integral image, and use this to calculate the simple feature based on subtracting the light-gray region from the dark-gray region. Explain how such pattern can be used to generate multiple simple features in the given image patch.

90	95	91
85	120	100
50	70	60

Figure 11

[10 Marks]

b. As discussed in the lecture, a convolutional filter can be understood as a shared template with respect to the input image or input feature map. Figure 12 shows a list of 64 learned filters of the first convolutional layer of AlexNet pretrained on ImageNet. Discuss how these filters work. Elaborate your discussion with respect to filters 4, 37, and 59, respectively.

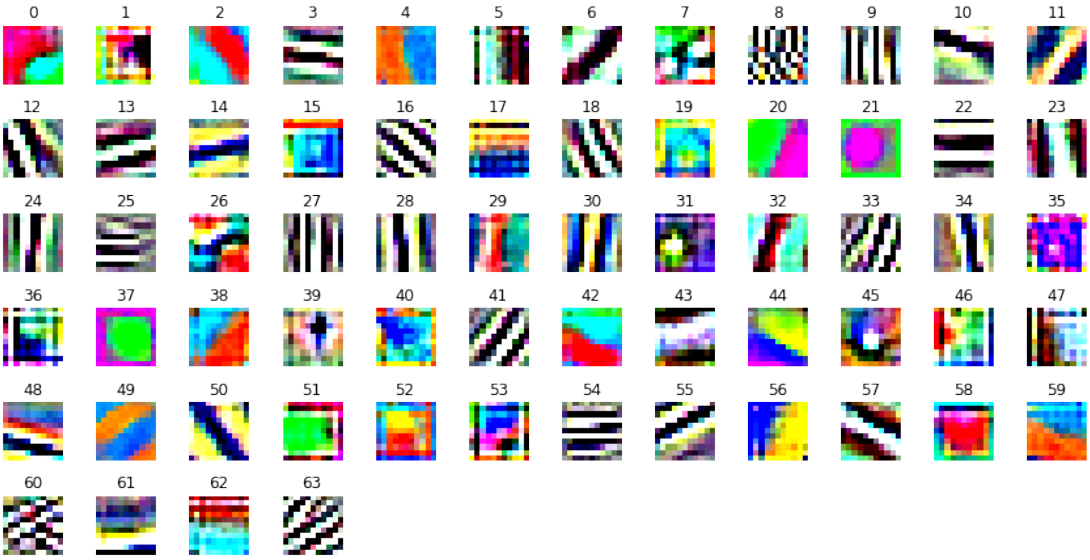


Figure 12

[10 Marks]

c. Both Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) are generative models that can generate realistic synthetic images. Compare and contrast the principles of VAEs and GANs, and their strength and weaknesses.

[10 Marks]